Customer Retention Enhancement Through Predictive Analytics

Machine Learning Model Development and Evaluation

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1. Introduction

The objective of Phase 2 was to develop a machine learning model capable of predicting customer churn using the cleaned and preprocessed dataset generated during Phase 1. Predicting customer churn enables banks to identify customers at risk of leaving and implement proactive retention strategies.

The cleaned dataset was used to train and evaluate a classification model to determine whether a customer is likely to churn or remain with the bank.

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2. Machine Learning Approach

Customer churn prediction is a binary classification problem where:

- Churn Status = 0 → Customer Retained
- Churn Status = 1 → Customer Churned

The following features were used as input variables:

- Age
- Gender
- Marital Status
- Income Level
- Amount Spent
- Login Frequency
- Service Usage
- Resolution Status

Customer ID was excluded from model training because it serves only as a unique identifier and does not contribute to prediction.

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3. Model Selection

Several classification algorithms can be used for churn prediction, including Logistic Regression, Decision Trees, Support Vector Machines, and Random Forest.

For this project, a **Random Forest Classifier** was selected due to:

- Ability to handle complex relationships between variables
- Reduced risk of overfitting
- Strong predictive performance
- Capability to identify important features influencing churn

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4. Data Splitting

The dataset was divided into training and testing datasets.

- Training Data: 80%
- Testing Data: 20%

The training dataset was used to train the model, while the testing dataset was used to evaluate its performance on unseen data.

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5. Model Training

The Random Forest Classifier was trained using the prepared customer dataset. During training, the model learned patterns and relationships between customer attributes and churn behaviour.

The trained model was then used to predict churn outcomes for the testing dataset.

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6. Model Evaluation

Accuracy

Model Accuracy: **75.5%**

```
#Calculate Accuracy
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.755
```

Interpretation

The model correctly predicted customer churn status for approximately 75.5% of customers in the testing dataset.

Classification Report

```
#Classification Report
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.75      1.00      0.86       150
           1       1.00      0.02      0.04        50

    accuracy                           0.76       200
   macro avg       0.88      0.51      0.45       200
weighted avg       0.82      0.76      0.65       200
```

Interpretation

The classification report provides Precision, Recall, and F1-Score for each class.

Results obtained:

Class 0 (Retained Customers)

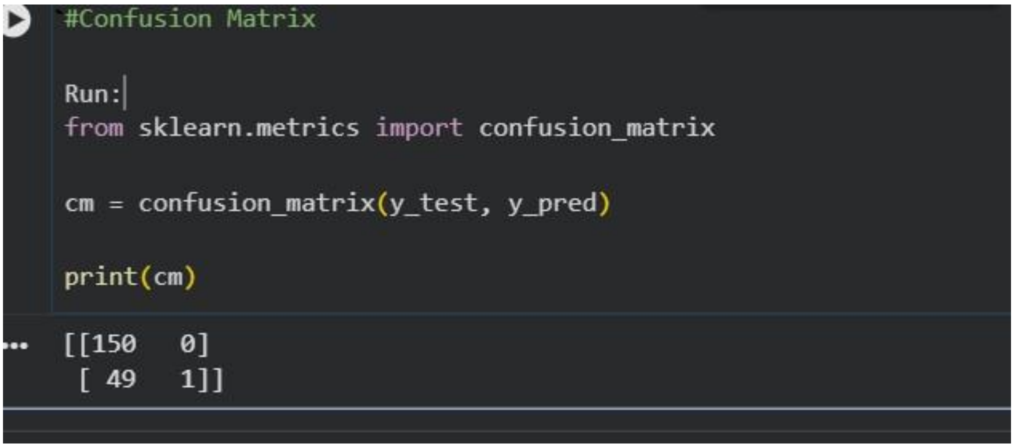
- Precision: 0.75
- Recall: 1.00
- F1-Score: 0.86

Class 1 (Churned Customers)

- Precision: 1.00
- Recall: 0.02
- F1-Score: 0.04

The model performed very well in identifying retained customers but struggled to correctly identify churned customers.

Confusion Matrix



```
#Confusion Matrix

Run:
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

print(cm)
```

```
[[150   0]
 [ 49   1]]
```

Confusion Matrix:

[[150, 0],
[49, 1]]

Interpretation

The confusion matrix shows:

- 150 retained customers correctly classified
- 1 churned customer correctly classified
- 49 churned customers incorrectly classified as retained

This indicates that the model has difficulty identifying churned customers due to class imbalance within the dataset.

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7. Feature Importance Analysis

[Insert Feature Importance Table Screenshot]

[Insert Feature Importance Graph Screenshot]

Interpretation

Feature importance analysis was performed to determine which variables contributed most significantly to customer churn prediction.

The most influential features can be used by the bank to design targeted customer retention strategies and improve customer engagement.

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8. Key Findings

- Random Forest Classifier successfully predicted customer churn with an overall accuracy of 75.5%.
- The model performed strongly for retained customers.

- Churn prediction performance was affected by class imbalance.
- Customer engagement and behavioural features contributed significantly to model predictions.
- Feature importance analysis provided insights into the factors influencing churn.

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9. Recommendations

To improve model performance in future iterations:

- Apply SMOTE (Synthetic Minority Oversampling Technique) to balance the dataset.
- Perform hyperparameter tuning using Grid Search or Random Search.
- Experiment with advanced algorithms such as XGBoost and Gradient Boosting.
- Introduce additional customer behaviour features for improved prediction accuracy.

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10. Conclusion

A Random Forest Classifier was successfully developed and evaluated for customer churn prediction. The model achieved an accuracy of 75.5% and demonstrated strong performance in identifying retained customers.

Although the model showed limitations in detecting churned customers due to class imbalance, it provides a solid foundation for customer retention analytics. Further improvements through data balancing and model optimization can enhance predictive performance and support more effective business decision-making.